NEELAY SHAH

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Website ◇ GitHub ◇ LinkedIn ◇ Google Scholar

EDUCATION

Birla Institute of Technology and Science Pilani, Goa, India 2018 - Present
Bachelor of Engineering (B.E. Hons.), Electronics and Instrumentation

EXPERIENCE

Biologically Inspired Neural Network Labs, BITS Pilani Goa 2020
Mentor: Basabdatta Sen Bhattacharya
In collaboration with TCS Research, India

- Implemented biologically plausible algorithms of speech recognition using optimized neuronal network simulation software
- Worked on extending current spiking neural network (SNN) based speech classification architectures with limited functionality to perform large vocabulary automatic speech recognition

Centre of Intelligent Robotics, IIIT Allahabad 2020
Intern (remote) | Mentor: GC Nandi

- Used model compression techniques to develop compact multi-modal emotion recognition models using deep learning for human-robot interaction
- Implemented generative grasping convolutional neural networks (GGCNN) for robotic grasping

Aurum Smart Tech 2020
Intern (remote)

- Developed an image processing pipeline for QR code scanning

Digital Communication Lab, BITS Pilani Goa 2020
Mentor: Abhijit Dey

- Developed a deep learning pipeline to forecast irregularities in the ionosphere which cause fluctuations in satellite communication signals
- Benchmarked machine learning models for classifying multi-path interference in satellite signals

PUBLICATIONS/PREPRINTS

KD-Lib: A PyTorch library for Knowledge Distillation, Pruning and Quantization
Het Shah, Avishree Khare, ***Neelay Shah****, Khizir Siddiqui*
arXiv preprint
[Paper] [Code]

TECHNICAL SKILLS

Programming Languages	Python, C++, Java
Frameworks/Libraries	PyTorch, Keras, PyGeNN, scikit-learn, JAX, NumPy, Pandas, Matplotlib
Operating Systems	Linux, Windows
Tools	Git/GitHub, LaTeX, Vim

OPEN-SOURCE SOFTWARE

KD-Lib
[Code] [Documentation]

- An open-source PyTorch library for making model compression research in deep learning more accessible
- Contains implementations of algorithms from 3 families of model compression research - Knowledge Distillation, Pruning, and Quantization
- Simple and intuitive API to facilitate the use of compression techniques for custom deep learning models with few lines of code

PERSONAL PROJECTS

Transfer Learning Aided One-shot Classification

- Transfer learning to improve the performance of a Siamese neural network for one-shot recognition on the MNIST dataset

Deep Learning Optimizers

- Toy implementations of popular deep learning optimization techniques from scratch

Machine Learning for Network Forensics

- Machine learning based network forensic mechanism for botnet activities in the Internet of Things (IoT)

GANs for Robotic Grasp Generation

- Exploratory use of image-to-image translation generative adversarial networks (GANs) to generate grasp orientations for robotic manipulation

Signboard Detection

- Detection of signboards containing text in vernacular languages using deep learning

RELEVANT COURSEWORK

University	Calculus, Linear Algebra, Probability and Statistics, Discrete Mathematics, Digital Design, Introduction to Cognitive Neuroscience, Signals and Systems, Control Systems, Data Structures and Algorithms*, Object-oriented Programming*
Online	Convolutional Neural Networks for Visual Recognition (Stanford CS231n), Machine Learning (Stanford CS229), Introduction to Reinforcement Learning (UCL)

* = ongoing

MENTORSHIP

Instructor - Introduction to Deep Learning
Student-run course
Centre for Technical Education, BITS Pilani Goa

- Taught and created reference material for an introductory deep learning course intended for college freshmen

OTHER ACTIVITIES

Core member
Society for Artificial Intelligence and Deep Learning

- Involved in reading and discussion sessions on artificial intelligence research

Member
University cricket team

· Participated in university level cricket tournaments

MISCELLANEOUS

Languages spoken	English, Hindi, Gujarati, Marathi, German (A1 level)
Interests	Sports (esp. Cricket and Football), International Relations, Journalism